

K2-30b

R-2319

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_211027 · created 2026-08-18T00:50:43+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459514.6888 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.013
Transit depth (Rp/Rs)^2	2.67 +/- 0.44 [%]
Orbital Inclination (inc)	88.42 +/- 0.87
Airmass coefficient 1 (a1)	1.151 +/- 0.031
Airmass coefficient 2 (a2)	-0.1175 +/- 0.0064
Scatter in the residuals of the lightcurve fit is	2.63 %
Best Comparison Star	None
Optimal Aperture	3.6
Optimal Annulus	7.95
Transit Duration (day)	0.1085 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.013 vs 0.1266 (deviation 2.78σ)
Duration fit vs published	0.1085 d vs 0.0971 d (deviation 2.02σ)
Mid-time O–C	-11.2 min
Depth SNR	12.4
Residual scatter	2.63 %

Light curve

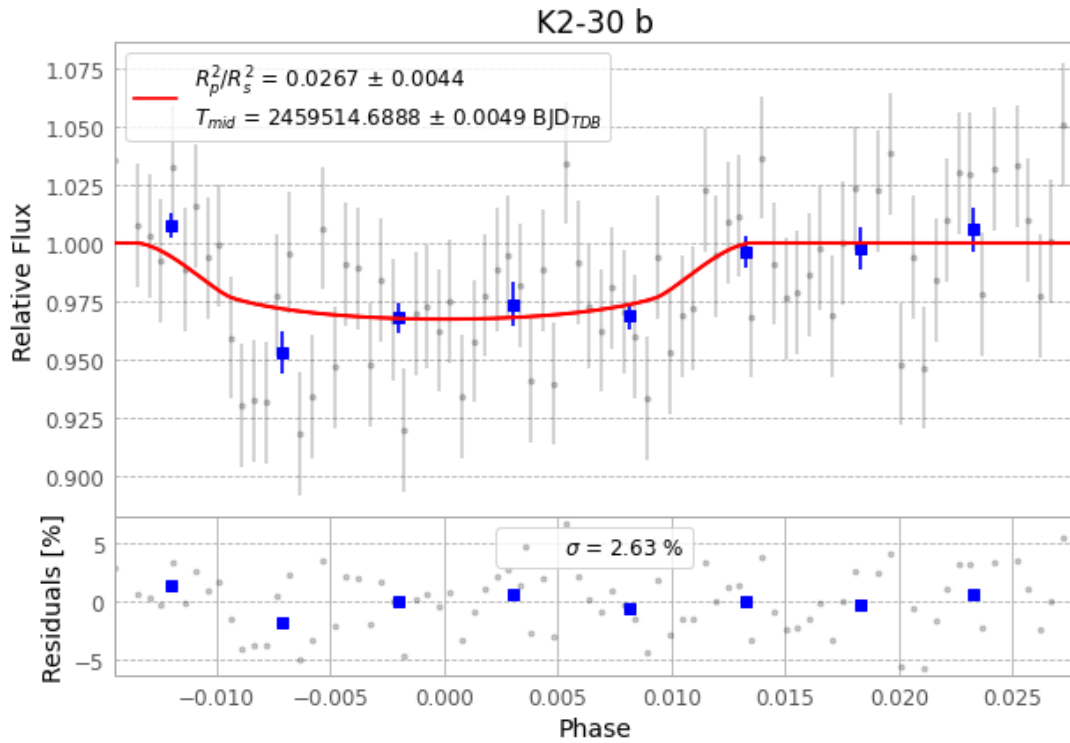


Figure 1 — FinalLightCurve_K2-30 b_2021-10-26.png (SHA-256 c3b0efd9a5ba943c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [399, 387], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b9fb36e5ff45e7ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

83 FITS frames (55 MB), K2-30211027025710.FITS → K2-30211027070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-211027.log (d5603b008c68...), FinalLightCurve_K2-30 b_2021-10-26.png (c3b0efd9a5ba...), FinalLightCurve_K2-30 b_2021-10-26.csv (09125a05d051...)

Compute resources

Wall time 107.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:50Z.